

1. Figure CQ19.1 shows a positive and a negative charge each moving in the xy -plane through three different magnetic fields. In each case, find the direction of the magnetic force on the charged particle. Indicate your answers as $+x$ or $-x$ for the positive or negative x -direction and similarly for the y - and z -directions.

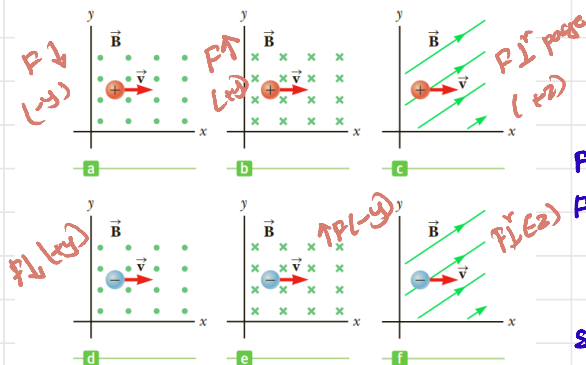


Figure CQ19.1

Dots = B out of the page
Crosses = B into the page.

Magnetic force on a moving charge:

$$\vec{F} = q\vec{v} \times \vec{B}$$

Right hand rule

For positive charge $\rightarrow$ use RHR
For negative charge $\rightarrow$ reverse the direction.

Steps for RHR:

1. Point finger $\rightarrow$ velocity $\vec{v}$
2. Curl toward $\rightarrow$ Magnetic field B .
3. Thumb $\rightarrow$ Force direction

2. Which way would a compass point if you were at Earth's north magnetic pole?

- The field is vertical
- There is no horizontal direction to align with.

It would not point in any direction, it will try to dip vertically downward.

3. How can the motion of a charged particle be used to distinguish between a magnetic field and an electric field in a certain region? Give a specific example to justify your answer.

Core idea:

Electric field: A charge feels a force even if it is at rest.

$$\vec{F} = q\vec{E}$$

Magnetic field: A charge feels a force only if it is moving.

$$\vec{F} = q\vec{v} \times \vec{B}$$

The distinguishing observation

If the particle speeds up or slows down $\rightarrow$ Electric field.
(force is along motion $\rightarrow$ work is done $\rightarrow$ KE changes)

If the particle moves in a circle at a constant speed $\rightarrow$ Magnetic field.
(force $\perp$ velocity $\rightarrow$ no work $\rightarrow$ speed constant).

4. Can a constant magnetic field set a proton at rest into motion?
Explain your answer.

$$\vec{F} = q\vec{v} \times \vec{B}$$

If the proton is at rest! $v=0$

$$\vec{F} = q(0) \times \vec{B} = 0$$

No force $\rightarrow$ No acceleration $\rightarrow$ No motion

Magnetic fields can deflect moving charges, but they cannot start moving them.

5. The following statements are related to the force of a magnetic field on a current-carrying wire. Indicate whether each statement is true (T) or false (F). (a) The magnetic force on the wire is independent of the direction of the current.

(b) The force on the wire is directed perpendicular to both the wire and the magnetic field. (c) The force takes its largest value when the magnetic field is parallel to the wire.

$$\vec{F} = q\vec{v} \times \vec{B}$$

The force is always perpendicular to the velocity.

A perpendicular force:

- changes direction
- does not change speed.

so:

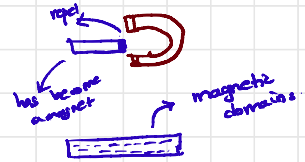
- motion becomes curved / circular
- magnitude of velocity stays the same.

(a) False

(b) True

(c) False.

6. Will a nail be attracted to either pole of a magnet? Explain what is happening inside the nail when it is placed near the magnet.



A nail is made of iron $\rightarrow$ ferromagnetic material.

The nail is attracted to the north and south pole.